

Ques. HOW CAN I INTERFACE A LASER DIODE WITH ARDUINO UNO? PLEASE LIST THE DIFFERENT TYPES OF LASERS AVAILABLE IN THE MARKET?

Pamarthi Kanakaraja

Ans. There are various ways to interface a laser with Arduino depending on the application. A laser diode finds applications in CD/DVD reading and recording, optical-fibre communications, laser printing and scanning, laser pointers, directional lighting, barcode readers and so on. A laser diode can be interfaced with Arduino



Fig. 1: A laser pointer



Fig. 2: KY-008 laser module

Uno using photoresistors, photodiodes, phototransistors or similar sensors and detectors.

Laser diodes and modules are available in various sizes and shapes. For example, a 3V to 5V DC, 5mW laser module is available in a metal case (Fig. 1). It produces a laser of about 650nm wavelength and a tiny red dot. It comes with three 1.5V button cells. Remove the cells and connect two wires: the positive wire to its body and the negative to the spring inside it. Connect the positive terminal to any one of the digital pins of Arduino board, and the negative wire to GND pin. Write a simple sketch (software) to turn on/off the laser torch through Arduino board.

Laser modules like KY-008 (Fig. 2) can also be connected directly to Arduino digital pins. This module can draw a current up to 30mA only,

so it can be directly powered from an Arduino board.

You may refer to the use of KY-008 laser module with Arduino from the following link: <http://henrysbench.capt.fatz.com/henrys-bench/arduino-output-devices/ky-008-arduino-laser-module-guide-and-tutorial/>

This link gives not only pin connection details but also Arduino code. You can also interface SF02 module directly with Arduino. SF02 is a lightweight laser rangefinder used for distance measurements ranging from 0 to 50 metres. Another interesting laser module from STMicroelectronics is VL53L0X. It works off 2.6-3.5V operating voltage and has I2C interface.

Types of lasers. There are many types of lasers available for various applications. Depending on the lasing medium used, these are solid-state, gas, excimer, dye or semiconductor lasers.

- Solid state lasers emit infrared light at 1.064 micrometres.
- Gas lasers have a primary output of a visible red light. CO₂ lasers emit energy in the far-infrared 10.6 micrometres, and are used for cutting hard materials.
- Excimer lasers produce light in the ultraviolet range.
- Dye lasers are tunable over a broad range of wavelengths.
- Semiconductor lasers are generally very small and use low power. These are used as the writing source in some laser printers, CD/DVD players and the like.

Q2. WHAT LANGUAGES CAN BE USED FOR ARDUINO UNO AND RASPBERRY PI?

Pournima P.R.

A2. C/C++, Python, Java, JavaScript and Perl are some popular program-

ming languages for embedded systems, Arduino Uno and Raspberry Pi. According to a recent study, more than 95 per cent of programming languages used in embedded system today are written in C or C++.

C programming language is currently the most popular for the Internet of Things (IoT) and embedded system programming. It is because with C you have easy access to physical hardware, good control over memory management and better performance as compared to other programming languages.

According to another report, the fastest-growing programming language for embedded computing is Python.

Some Arduino and Raspberry Pi projects may be highly compact, so simple program environments or languages like B# (or B sharp) are more suitable.

Arduino Uno. Arduino Uno works on Arduino integrated development environment (IDE), which supports C and C++ using special rules of code structuring, based on Wiring Project. A program written with Arduino IDE is called a sketch. There are also other programming languages in use for Arduino projects, like Python, Ardublock and Snap4Arduino.

Raspberry Pi. GCC, Scratch, Java and Ruby come pre-installed on Raspberry Pi. The Raspberry Pi Foundation recommends Python and Scratch as the programming languages because these are easier to pick up. However, any language that can compile for its processor like ARMv6 can be used with Raspberry Pi.

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